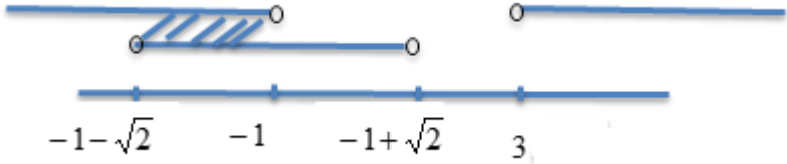


ANSWER SCHEME: MUAFAKAT SET 3 SM015

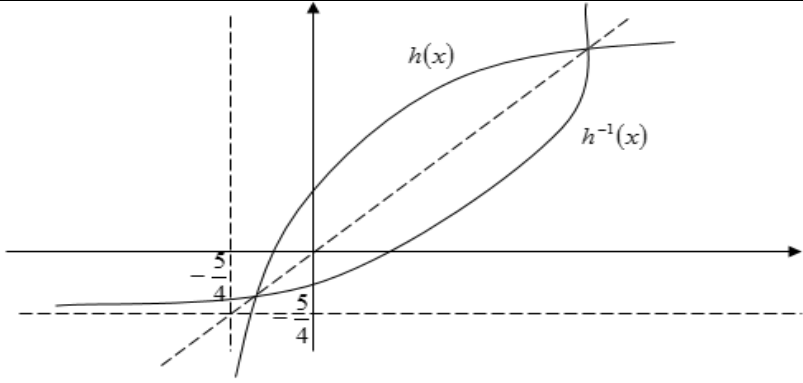
Q	SOLUTION	MARK
SECTION A		
1 (a)	$\lim_{x \rightarrow 1^-} \frac{x^2 + x - 2}{ x - 1 } = \lim_{x \rightarrow 1^-} \frac{x^2 + x - 2}{-(x - 1)}$ $= \lim_{x \rightarrow 1^-} \frac{(x + 2)(x - 1)}{-(x - 1)}$ $= \lim_{x \rightarrow 1^-} -(x + 2) = -(1 + 2)$ $= -3$	K1 K1 K1 J1
1(b)	$\lim_{x \rightarrow +\infty} \sqrt{x}(\sqrt{x+1} - \sqrt{x}) = \lim_{x \rightarrow +\infty} (\sqrt{x(x+1)} - x) \times \frac{(\sqrt{x(x+1)} + x)}{(\sqrt{x(x+1)} + x)}$ $= \lim_{x \rightarrow +\infty} \frac{x(x+1) - x^2}{\sqrt{x(x+1)} + x}$ $= \lim_{x \rightarrow +\infty} \frac{x^2 + x - x^2}{\sqrt{x(x+1)} + x}$ $= \lim_{x \rightarrow +\infty} \frac{x}{\sqrt{x(x+1)} + x}$ $= \lim_{x \rightarrow +\infty} \frac{\frac{x}{x}}{\frac{\sqrt{x^2 + x} + \frac{x}{x}}{\frac{x}{x}}}$ $= \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1}$ $= \frac{1}{2}$	K1 K1 K1 K1 J1
	TOTAL	9
2(a)	$y = \sin(x^2 + 1)$ $\frac{dy}{dx} = 2x \cos(x^2 + 1)$ $u = 2x \qquad v = \cos(x^2 + 1)$ $u' = 2 \qquad v' = -2x \sin(x^2 + 1)$	K1 K1

	$\frac{d^2 y}{dx^2} = -4x^2 \sin(x^2 + 1) + 2 \cos(x^2 + 1)$ $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = x[-4x^2 \sin(x^2 + 1) + 2 \cos(x^2 + 1)]$ $- [2x \cos(x^2 + 1)] + 4x^3 \sin(x^2 + 1)$ $= -4x^3 \sin(x^2 + 1) + 2x \cos(x^2 + 1)$ $- 2x \cos(x^2 + 1) + 4x^3 \sin(x^2 + 1)$ $= 0$	K1 K1 J1
2(b)(i)	$y = (\ln x)^6$ $\frac{dy}{dx} = 6(\ln x)^5 \left(\frac{1}{x}\right)$ $= \frac{6}{x} (\ln x)^5$	K1 J1
2(b)(ii)	$y = e^{3x}(2x^4 + 1)$ $u = e^{3x} \quad v = 2x^4 + 1$ $u' = 3e^{3x} \quad v' = 8x^3$ $\frac{dy}{dx} = 8x^3 e^{3x} + 3e^{3x}(2x^4 + 1)$ $= e^{3x}(8x^3 + 6x^4 + 3)$	K1 K1 J1
	TOTAL	10
3	$f(x) = x^3 - 3x^2 - 9x + 11$ $f'(x) = 3x^2 - 6x - 9$ $f'(x) = 0$ $3x^2 - 6x - 9 = 0$ $(x - 3)(x + 1) = 0$ $x = 3, x = -1$ When $x = 3$ $f(3) = -16$ When $x = -1$ $f(-1) = 16$ $\therefore$ critical points are (3, -16) and (-1, 16) $\frac{d^2 y}{dx^2} = 6x - 6$ At point (3, -16)	B1 K1 J1 K1

	$\frac{d^2y}{dx^2} = 6(3) - 6$ $= 12 > 0(\text{min})$ at point $(-1, 16)$ $\frac{d^2y}{dx^2} = 6(-1) - 6$ $= -12 < 0(\text{max})$ $\therefore$ minimum point is $(3, -16)$ and maximum point is $(-1, 16)$	K1 J1
	TOTAL	6
SECTION B		
1(a)	$\frac{1}{z_1} = k \overline{z_1}$ $\frac{1}{2+i} = k(2-i)$ $\frac{1}{2+i} = 2k - ki$ $\frac{1}{2+i} \left(\frac{1}{2-i} \right) = 2k - ki$ $\frac{2}{5} - \frac{i}{5} = 2k - ki$ $\frac{2}{5} = 2k$ $k = \frac{1}{5}$	K1 K1 K1 J1
1(b)	$\frac{z_1^2}{z_1 - z_2} = \frac{(2+i)^2}{(2+i) - (1-2i)}$ $= \frac{5+4i}{3i+1}$ $= \left(\frac{5+4i}{1+3i} \right) \left(\frac{1-3i}{1-3i} \right)$ $= \frac{5-11i+12}{10}$ $= \frac{17}{10} - \frac{11}{10}i$	K1 J1 K1 K1 J1
	TOTAL	9
2(a)	$\log(x+2) + \log(x-1) = 1$ $\log(x+2) + \log(x-1) = \log 10$	K1

	$\log_{10}(x+2)(x-1) = \log_{10} 10$ $(x+2)(x-1) = 10$ $x^2 + x - 12 = 0$ $(x-3)(x+4) = 0$ $x = 3, x = -4$ check Sub $x = 3$ LHS : $\log(3+2) + \log(3-1) = \log 5 + \log 2$ $= 1$ Sub $x = -4$ RHS : $\log(-4+2) + \log(-4-1) = \log(-6) + \log(-5)$ (undefined) $\therefore x = 3$	K1 K1 K1 K1 J1
2(b)	$ 2 - x^2 < -2x - 1$ $2 - x^2 < -2x - 1$ $-x^2 + 2x + 3 < 0$ $x^2 - 2x - 3 > 0$ $(x-3)(x+1) > 0$ ZOF : $-1, 3$  $(-\infty, -1) \cup (3, \infty)$ AND $2 - x^2 > 2x + 1$ $-x^2 - 2x + 1 > 0$ $x^2 + 2x - 1 < 0$ ZOF : $-1 + \sqrt{2}, -1 - \sqrt{2}$  $(-1 - \sqrt{2}) \cup (-1 + \sqrt{2})$  $(-1 - \sqrt{2}, -1)$	B1 K1K1 K1 J1 K1 J1
	TOTAL	13

3(a)	Domain of function $h(x)$ $2 - 3x \geq 0$ $x \leq \frac{2}{3} \rightarrow \left(-\infty, \frac{2}{3}\right]$ Range of function $h(x)$: $[0, \infty)$	B1 B1
3(b)	$g \circ h(x) = \frac{1}{\sqrt{2-3x}}$ $h^2(x) = \sqrt{2-3(\sqrt{2-3x})}$ $g \circ h^2(x) = \frac{1}{\sqrt{2-3\sqrt{2-3x}}}$ $g \circ h^2(0) = \frac{1}{\sqrt{2-3\sqrt{2-3(0)}}} = \frac{1}{\sqrt{2-3\sqrt{2}}}$	K1J1 K1J1 K1J1
	TOTAL	8
4(a)	If $h(x) = f(2x+3)$, find $h(x)$ $h(x) = f(2x+3)$ $= \ln[2(2x+3)-1]$ $= \ln[4x+6-1]$ $= \ln 4x+5 $	K1 K1 J1
4(b)	$h[h^{-1}(x)] = x$ $\ln[4h^{-1}(x)+5] = x$ $4h^{-1}(x)+5 = e^x$ $4h^{-1}(x) = e^x - 5$ $h^{-1}(x) = \frac{1}{4}[e^x - 5]$ $R_{h^{-1}} = D_h = \left(-\frac{5}{4}, \infty\right)$	K1 K1 J1 J1

4(c)		R1 R1 R1
	TOTAL	10
5(a)	i) $f(1) = -2$ ii) $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{x^2 - 1}{1 - x} = \lim_{x \rightarrow 1^+} \frac{(x-1)(x+1)}{1-x} = \lim_{x \rightarrow 1^+} -(x+1) = -(1+1) = -2$ $\lim_{x \rightarrow 1^-} f(x) = -2$ $\therefore \lim_{x \rightarrow 1} f(x)$ exist because $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$ iii) $\lim_{x \rightarrow 1} f(x) = f(1) = -2$ $\therefore f(x)$ is continuous at $x = 1$	B1 K1 J1 K1 J1
5(b)	At $x = 3$ $\lim_{x \rightarrow 3^-} Ax^2 + 2 = \lim_{x \rightarrow 3^+} Bx + 1$ $A(3)^2 + 2 = B(3) + 1$ $9A + 2 = 3B + 1 \dots\dots\dots(1)$ At $x = 5$ $\lim_{x \rightarrow 5^+} 4 = \lim_{x \rightarrow 5^-} Bx + 1$ $4 = B(5) + 1$ $3 = 5B$ $B = \frac{3}{5}$ Solve (1) & (2) :	K1 K1 K1 K1 J1

7(b)	$\frac{dv}{dt} = 8cm^3s^{-1}$ $\frac{dh}{dt} = \frac{dh}{dv} \cdot \frac{dv}{dt}$ $V = \frac{4}{27}\pi h^3$ $\frac{dV}{dh} = \frac{4}{27}(3)\pi h^2$ $= \frac{4}{9}\pi h^2$ $\frac{dh}{dV} = \frac{9}{4\pi h^2}$ $\frac{dh}{dt} = \frac{9}{4\pi(12)^2} \cdot (8)$ $= \frac{72}{576\pi}$ $= \frac{1}{8\pi}$	K1 J1 K1 J1
	TOTAL	8